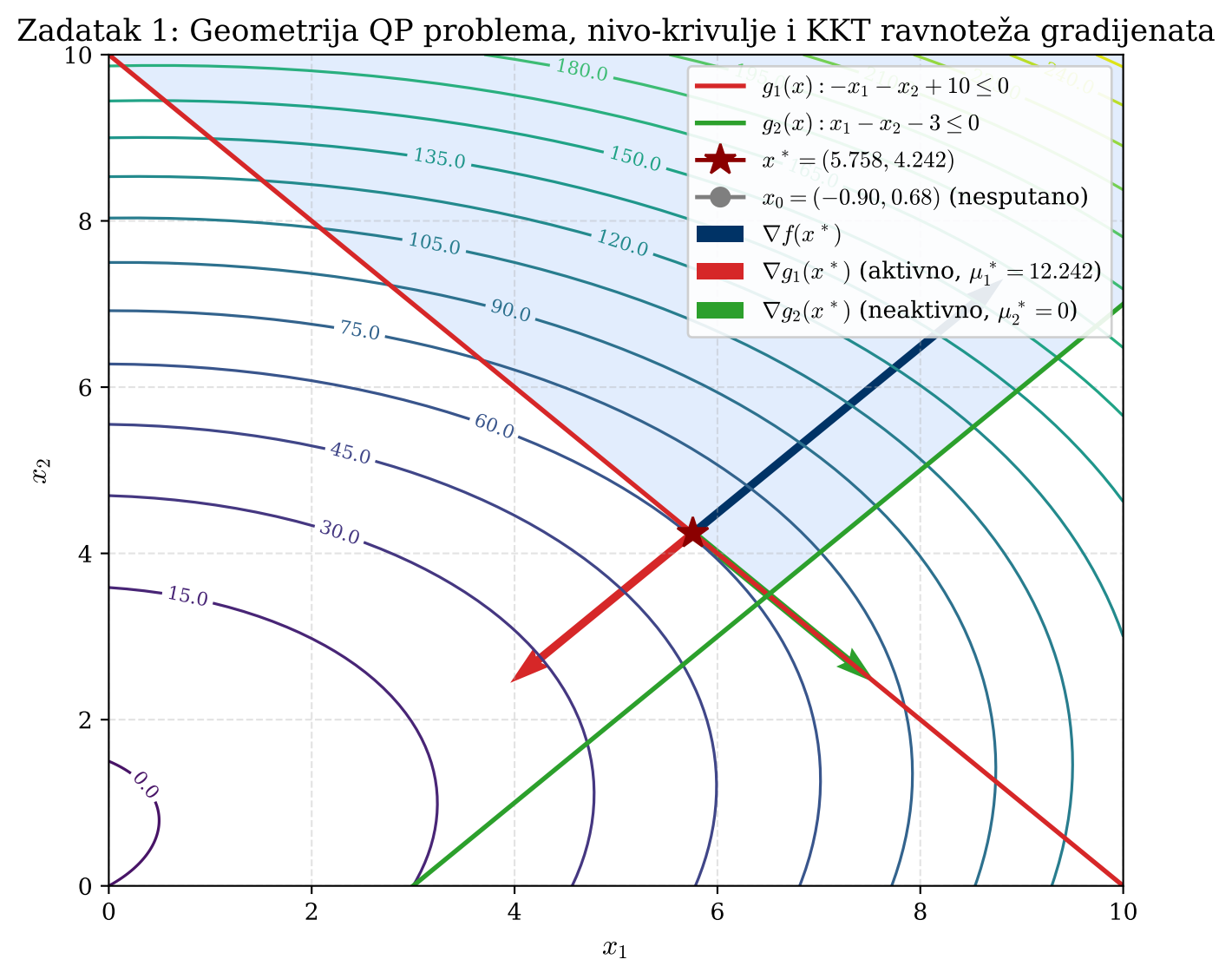
Zadatak 1: Geometrija QP problema, nivo-krivulje i KKT ravnoteža gradijenata
$g_1(x): -x_1 - x_2 + 10 \leq 0$
$g_2(x): x_1 - x_2 - 3 \leq 0$
$x^* = (5.758, 4.242)$
$x_0 = (-0.90, 0.68)$ (nesputano)
$\nabla f(x^*)$
$\nabla g_1(x^*)$ (aktivno, $\mu_1^* = 12.242$)
$\nabla g_2(x^*)$ (neaktivno, $\mu_2^* = 0$)
180.0
150.0
135.0
120.0
105.0
90.0
75.0
60.0
45.0
30.0
15.0
0.0
x_1
x_2
0
2
4
6
8
10